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Nonlinear Oscillator

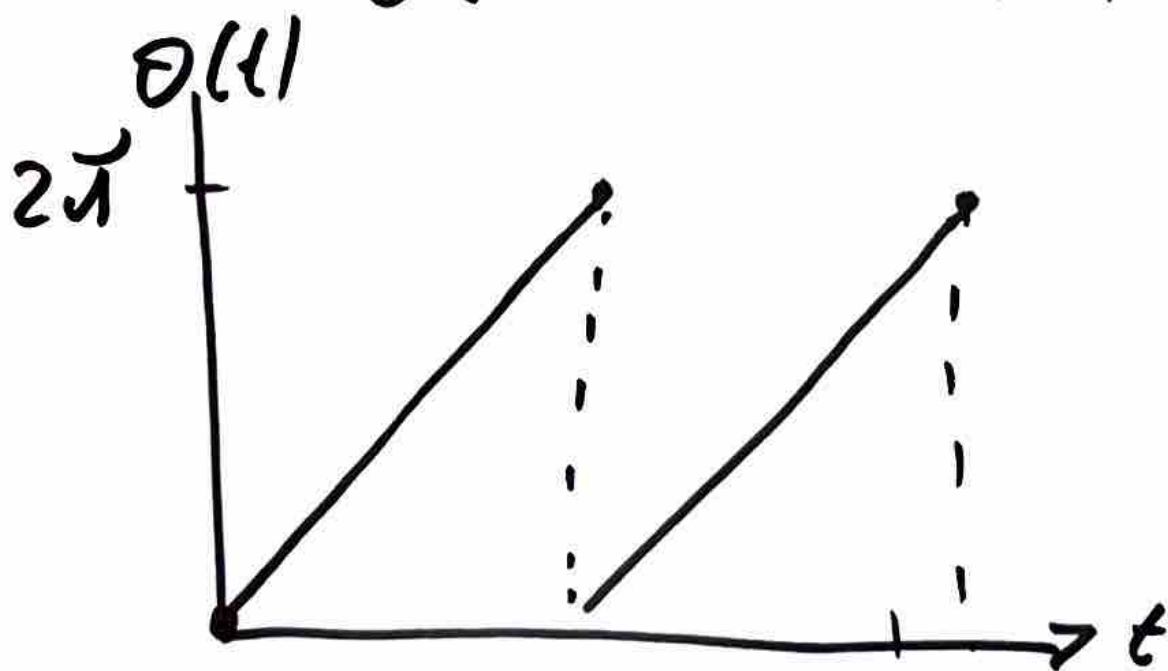
$$\frac{d}{dt} \theta(t) = \omega - \underbrace{a \sin \theta(t)}_{\text{nonlinear}}$$

$$\omega, a > 0$$

ω, a are two constants

$a = 0$: $\frac{d}{dt} \theta(t) = \omega$

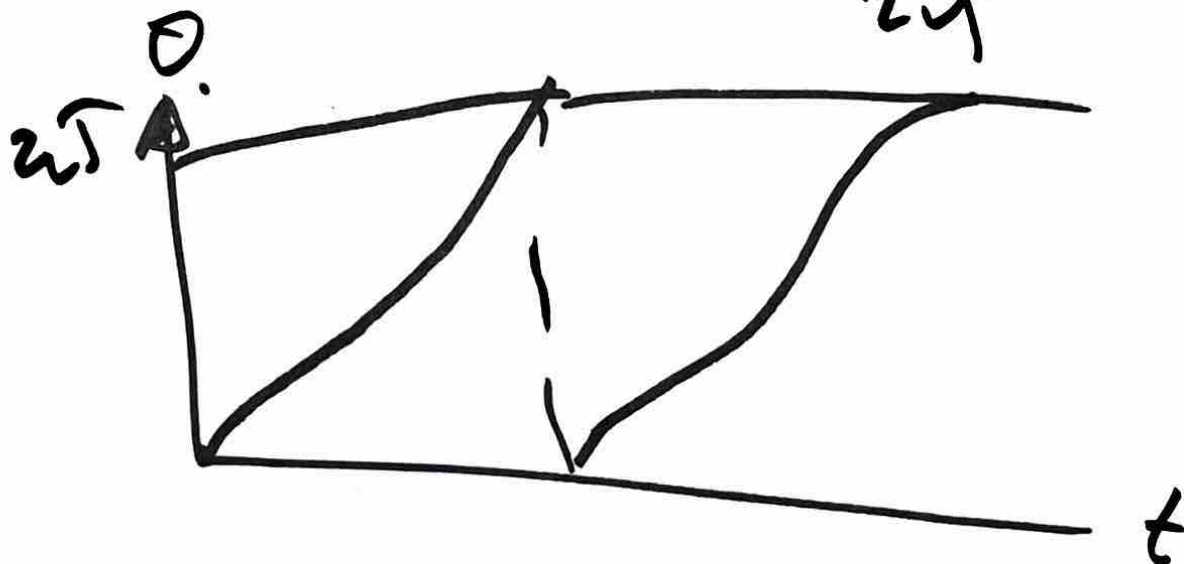
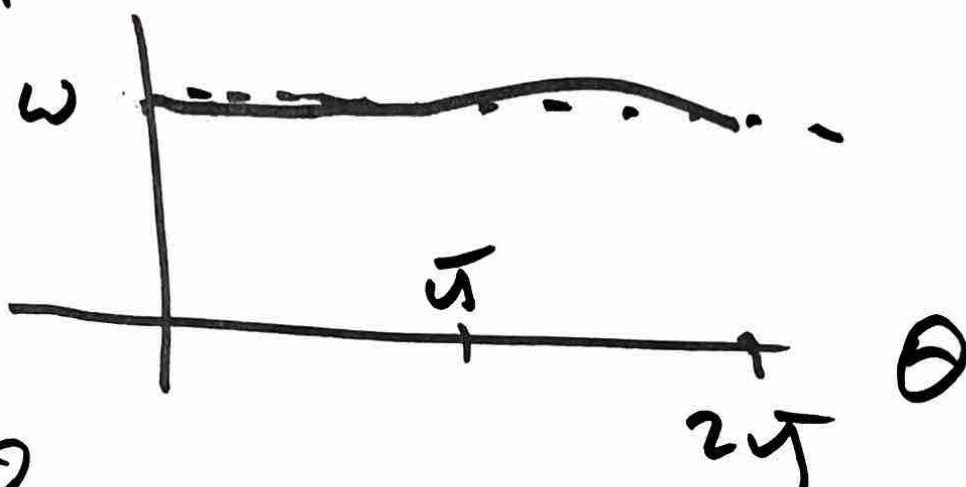
$$\theta(t) = \omega t + \theta(t=0)$$



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$$0 < a < \omega$$

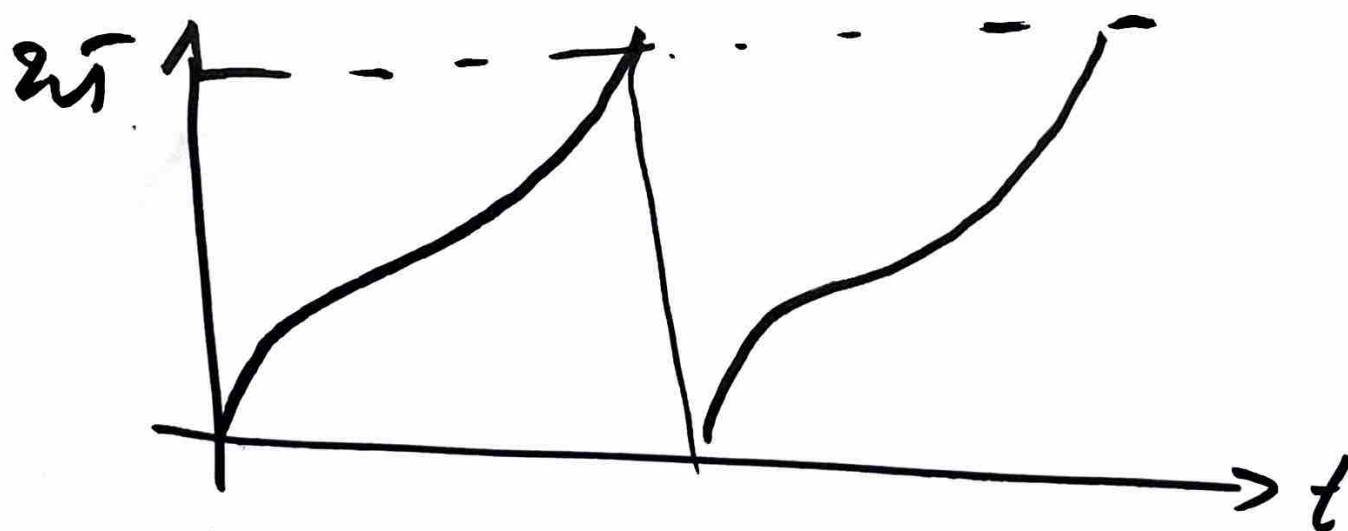
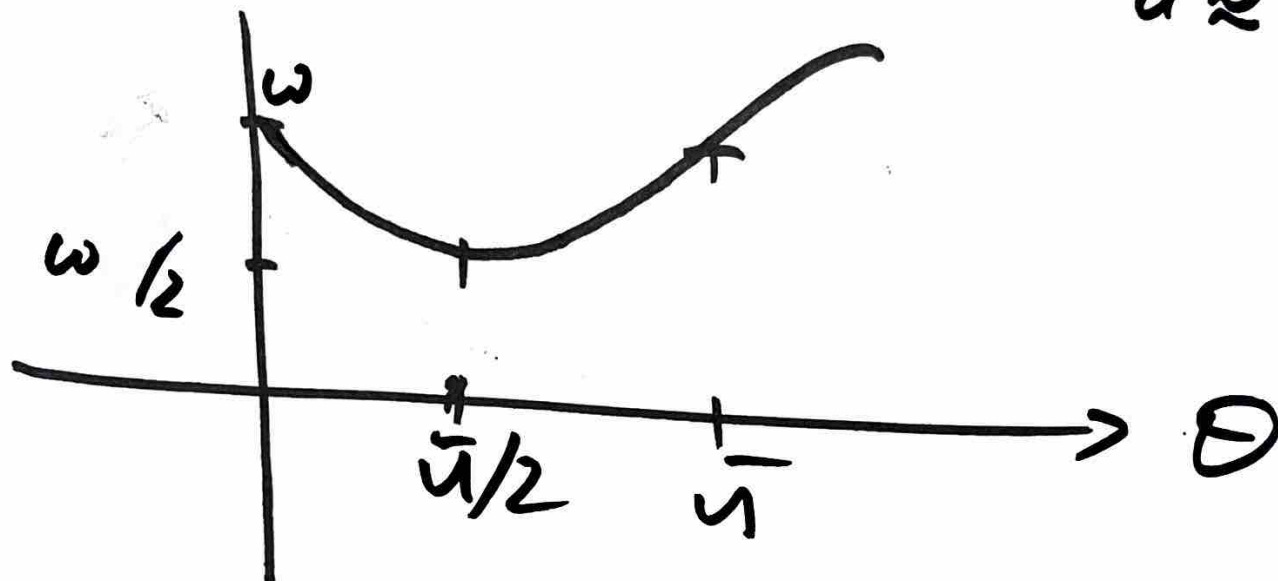
$$\frac{d}{dt} \Theta(t) = \omega - a \sin \Theta$$



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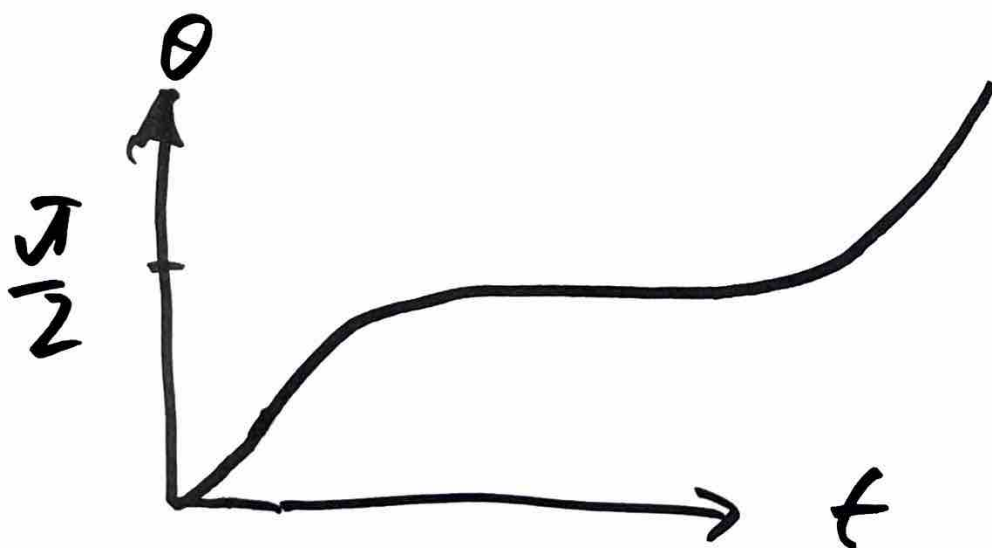
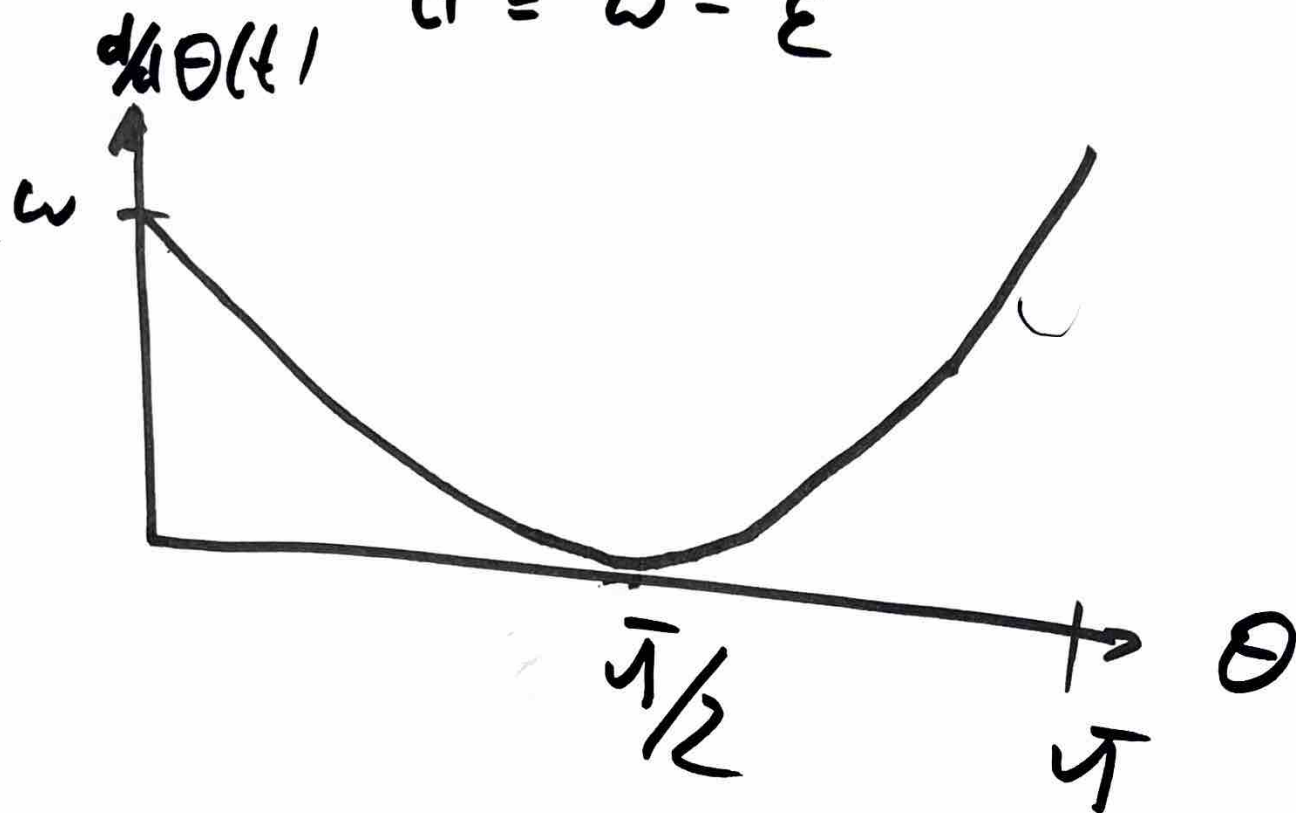
$$\dot{\Theta}(t) = \omega - a \sin \Theta$$

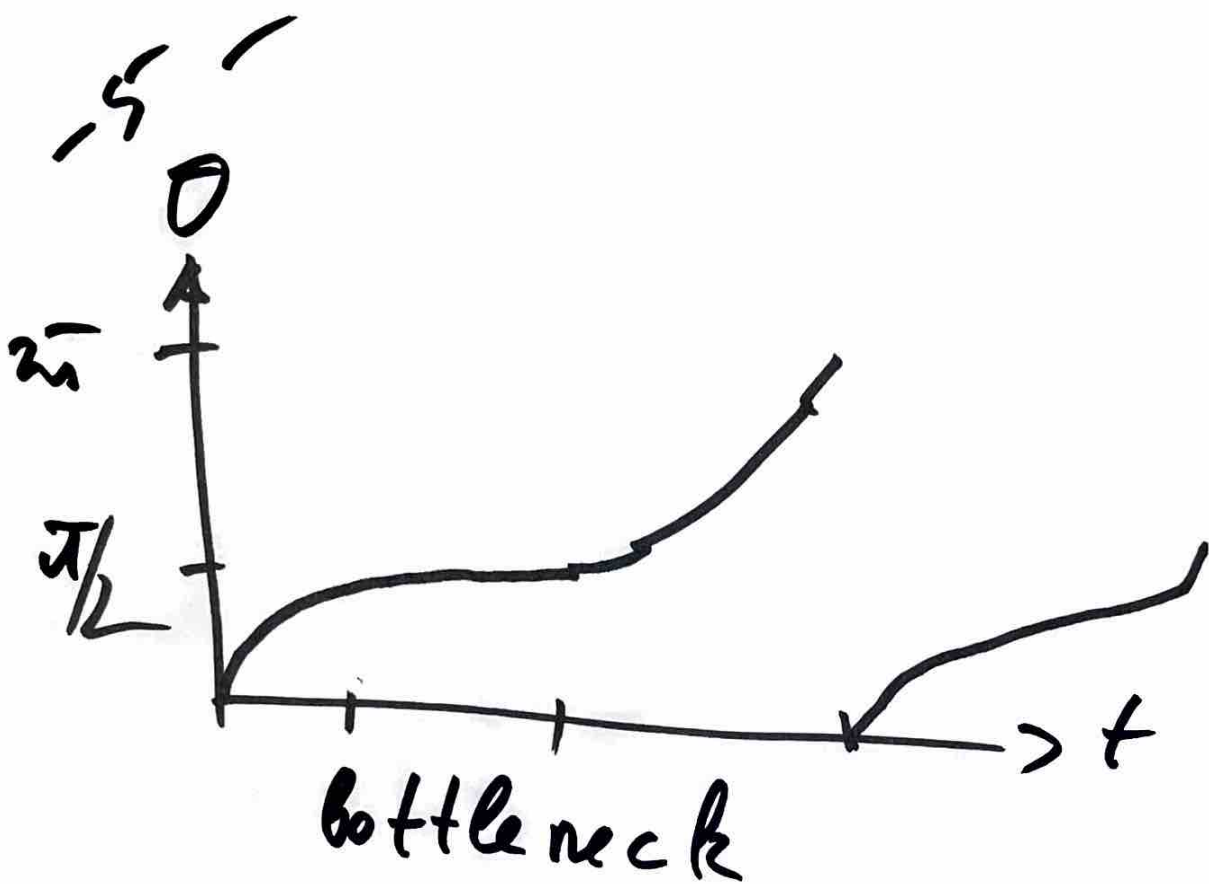
$\sigma < a < \omega$
 $a \approx \omega/2$



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$$\alpha = \omega - \varepsilon$$





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$$\frac{d}{dt} \theta(t) = \omega - a \sin \theta(t)$$

$$T = \int_0^T dt = \int_0^{2\pi} d\theta \cdot \frac{dt}{d\theta}$$

$$\frac{dt}{d\theta} = \frac{1}{\omega - a \sin \theta(t)}$$

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$$\frac{d\theta}{dt} = \omega - a \sin \theta,$$
$$\theta = \theta(t)$$

$$T = \int_0^T dt$$

$$= \int_0^{2\pi} d\theta \frac{dt}{d\theta} =$$

$$= \int_0^{2\pi} d\theta \cdot \frac{1}{\omega - a \sin \theta}$$

$$= \frac{2\pi}{\sqrt{\omega^2 - a^2}} = \frac{2\pi}{\sqrt{(\omega - a)(\omega + a)}}$$

Bottle neck

as $a \rightarrow \omega - \epsilon$

$$\lim_{a \rightarrow \omega - \epsilon} T(a) = \infty$$

$a \rightarrow \omega - \epsilon$

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$$\dot{\theta} = \omega - a \sin \theta$$

$$a = \omega - \varepsilon$$

$$\theta = \frac{\sqrt{t}}{2} + \mu x \quad \varepsilon, \mu \ll 1$$

$$\mu \dot{x} = \omega - (\omega - \varepsilon) \sin\left(\frac{\sqrt{t}}{2} + \mu x\right)$$

$$\sin\left(\frac{\sqrt{t}}{2} + x\right) = \cos x$$

$$\mu \dot{x} = \omega - (\omega - \varepsilon) \cos(\mu x)$$

$$= \omega - (\omega - \varepsilon) \left(1 - \frac{\mu^2 x^2}{2}\right)$$

$$\cos x \approx 1 - \frac{x^2}{2}, \quad |x| \ll 1$$

$$\mu \dot{x} = \underline{\omega} - \underline{\omega} + \omega \frac{\mu^2 x^2}{2} + \varepsilon - \varepsilon \frac{\mu^2 x^2}{2}$$

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$$J \dot{x} = \varepsilon + \frac{\omega J^2 x^2}{2} - \cancel{\varepsilon J^2 x^2 / 2}$$

$$\dot{x} = \frac{\varepsilon}{J} + \frac{\omega J}{2} x^2$$

Saddle node

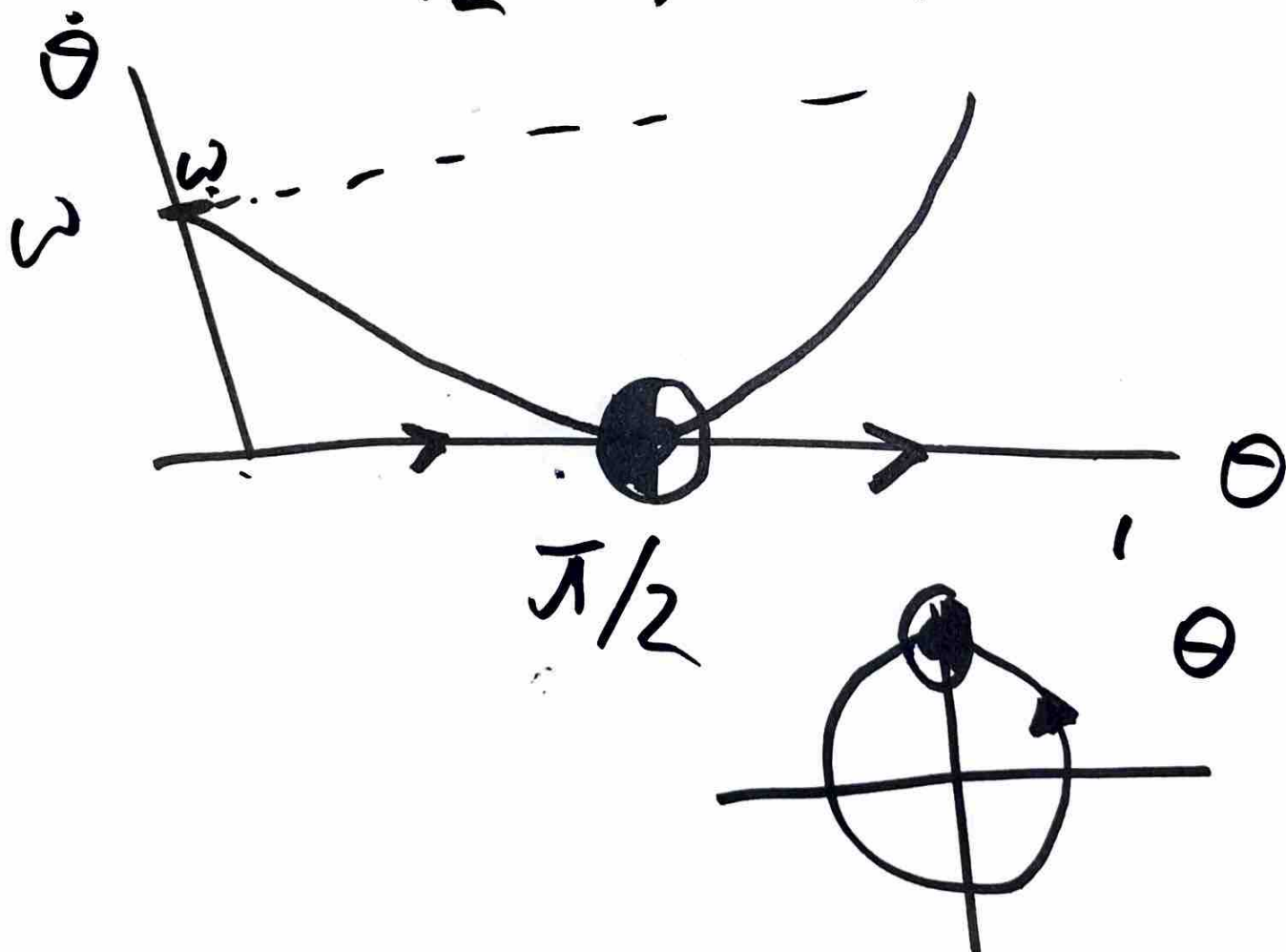
bifurcation

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$$\dot{\theta} = \omega$$

$$\dot{\theta} = \omega - \omega \sin \theta$$
$$= \omega(1 - \sin \theta)$$

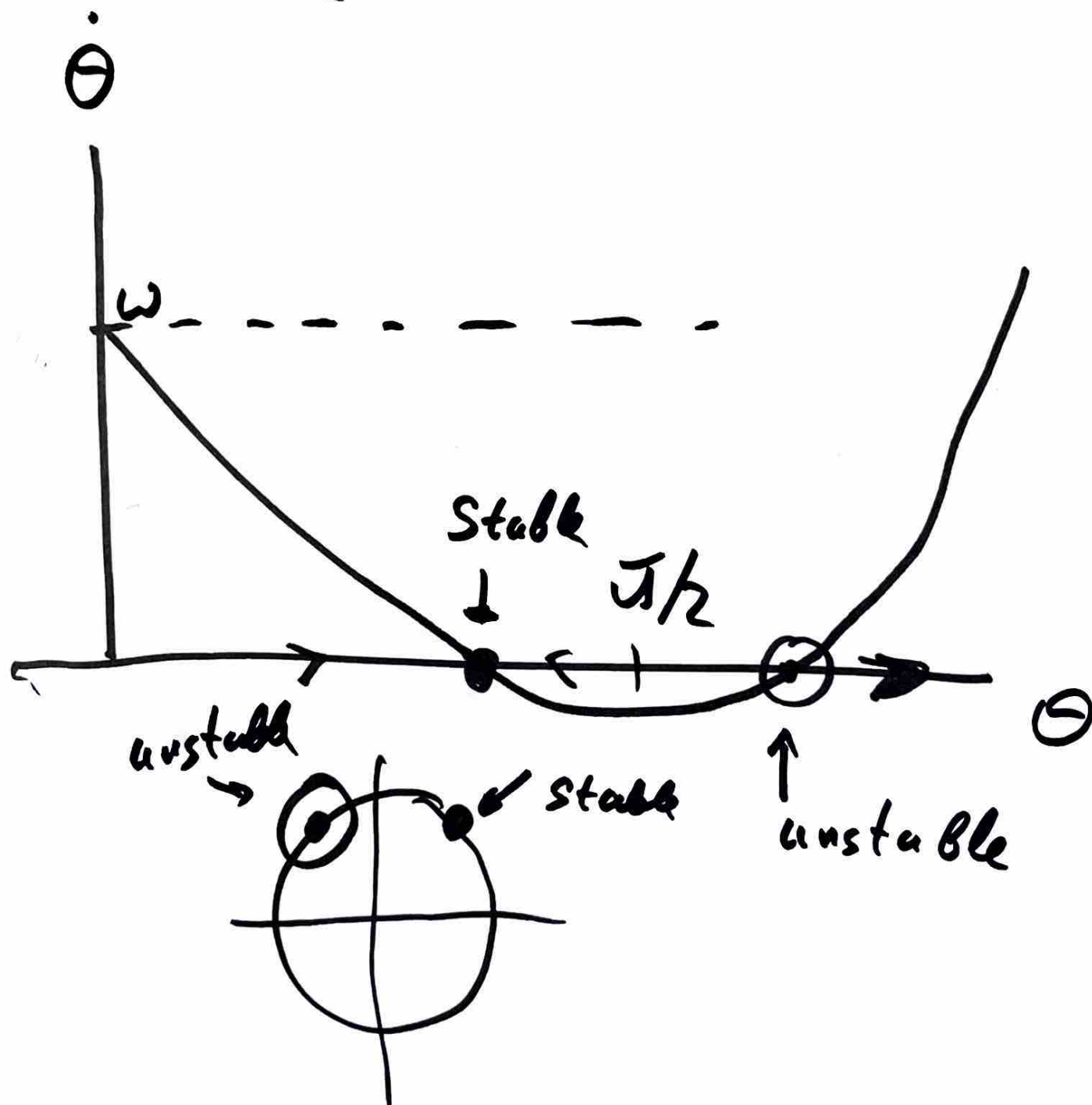
$$\theta^* = \pi/2 - \text{fixed point}$$



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$$\dot{\theta} = \omega - a \sin \theta$$

$$a = \omega + \varepsilon$$



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$$\frac{d}{dt} \theta(t) = f(\theta(t), a)$$

$$f(\theta, a) = \omega - a \sin \theta$$

$$\omega - a \sin \theta^* = 0$$

$$\sin \theta^* = \frac{\omega}{a}, \text{ exists only if } \omega/a < 1$$

$$\theta_1^* = \text{Arc Sin } \omega/a$$

$$a \geq \omega > 0$$

$$\theta_2^* = \pi - \text{Arc Sin } \omega/a$$

$$\sin(\pi - x) = \sin x$$

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$$\frac{\partial f(\theta, a)}{\partial \theta} = \frac{\partial}{\partial \theta} [\omega - a \sin \theta]$$

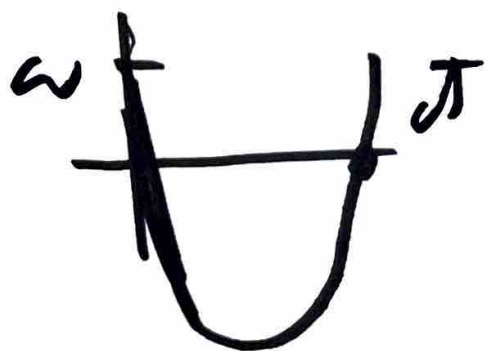
$$= -a \cos \theta$$

$$\theta^* = \text{Arc Sin } \omega/a ; \quad \sin \theta^* = \omega/a$$

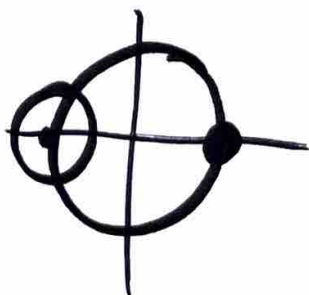
$$\cos \theta^* = \pm \sqrt{1 - \left(\frac{\omega}{a}\right)^2}$$

$$\dot{\theta} = \omega - a \sin \theta$$

$$a \gg \omega$$



$$\dot{\theta} \approx -a \sin \theta$$



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Pendulum

$$Lm \ddot{\theta}(t) + mg \sin \theta = 0$$

+ Add friction

$$mL\ddot{\theta} + \tilde{b}\dot{\theta} + mg \sin \theta = 0$$

times L

$$mL^2 \ddot{\theta} + L\tilde{b}\dot{\theta} + \overbrace{mgL \sin \theta}^{\text{torque}} = 0$$

Add torque

$$mL^2 \ddot{\theta} + L\tilde{b}\dot{\theta} + mgL \sin \theta = \tau$$

↑
external
torque

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$$mL^2 \ddot{\Theta} + b \dot{\Theta} + mgL \sin \Theta = \Gamma$$

$$[mgL] = \text{Mass} \cdot \text{Length}^2 / \text{Time}^2 \quad L\ddot{\Theta} = b$$

$$[mL^2 \ddot{\Theta}] = \text{Mass} \cdot \text{Length}^2 / \text{Time}^2$$

$$[b] = \text{Mass} \cdot \text{Length}^2 / \text{Time}$$

$$[\Gamma] = \text{Mass} \cdot \text{Length}^2 / \text{Time}^2$$

$$t = T \cdot \tau, \quad [t] = [T] = \text{Time}$$

$$[\tau] = 1$$

$$\frac{mL^2}{T^2} \frac{d^2}{d\tau^2} \Theta + \frac{b}{T} \frac{d}{d\tau} \Theta + mgL \sin \Theta = \Gamma$$

~~$$\frac{L}{T^2} \frac{d^2}{d\tau^2} \Theta(\tau) + \frac{b}{T} \frac{d}{d\tau} \Theta$$~~
~~$$T^2 g \frac{d^2}{d\tau^2} \Theta(\tau) + T mgL \sin \Theta$$~~

$$\frac{mL^2}{T^2} \frac{d^2}{dt^2} \theta + \frac{L}{T} \frac{d}{dt} \theta + mgL \sin \theta = L$$

$$\frac{L}{T^2 g} \frac{d^2}{dt^2} \theta + \frac{L}{T mgL} \frac{d}{dt} \theta + \sin \theta = \frac{L}{mgL}$$

$$\frac{L}{mgLT} = 1 \Rightarrow T = \frac{L}{mgL}$$

$$\frac{L^3 m^2 g}{L^2} \frac{d^2}{dt^2} \theta + \frac{d}{dt} \theta + \sin \theta = \frac{L}{mgL}$$

$$\frac{L^3 m^2 g}{L^2 g}$$

$$L = \frac{L^3 m^2 g}{L^2 g}$$

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$$\varepsilon = \frac{L^3 m^2 g}{b^2}$$

$$[\varepsilon] = \frac{L^3 m^2 L}{T^2 (M \cdot L^2 / T)^2} = 1$$

$$\varepsilon \ll 1, \quad \varepsilon \ddot{\theta} \rightarrow 0$$

$$\frac{d}{dt} \theta = -\sin \theta + \frac{\Gamma}{mgL}$$

$$\gamma = \frac{\Gamma}{mgL}$$

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$$\frac{d}{dt} \theta = -\sin \theta + \nu$$

• Zero torque: $\nu = 0$

$$\dot{\theta} = -\sin \theta$$

$\theta^* = 0$, stable

π unstable

tiny torque $\nu \gg 0 \Rightarrow \nu > 0$



• critical $\nu = 1$, 1 one fixed
mixed stability point

• Large torque $\nu > 1$ no fixed
points, periodic

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Fire flies.

Single fire fly

$\Theta(t)$ - light per time

$\Theta(t) = \emptyset$ - light

$\Theta(t) = \top$ - no light

$$\frac{d}{dt} \Theta(t) = \omega$$

fire fly alone

mimic
external

$$\frac{d}{dt} \Theta(t) = \omega + A \sin(\Theta - \psi(t))$$

$\psi(t)$ - external
signed

$$\frac{d}{dt} \psi(t) = \Omega$$

Ω external, period, τ